	OSI Layer	TCP/IP	Datagrams are called
Software	Layer 7 Application	HTTP, SMTP, IMAP, SNMP, POP3, FTP	Upper Layer Data
	Layer 6 Presentation	ASCII Characters, MPEG, SSL, TSL, Compression (Encryption & Decryption)	
	Layer 5 Session	NetBIOS, SAP, Handshaking connection	
	Layer 4 Transport	TCP, UDP	Segment
	Layer 3 Network	IPv4, IPv6, ICMP, IPSec, MPLS, ARP	Packet
Hardware	Layer 2 Data Link	Ethernet, 802.1x, PPP, ATM, Fiber Channel, MPLS, FDDI, MAC Addresses	Frame
	Layer 1 Physical	Cables, Connectors, Hubs (DLS, RS232, 10BaseT, 100BaseTX, ISDN, T1)	Bits

Figure 11.9 OSI LAYERS

are free of errors. This layer has simple protocols like "802.3 for Ethernet" and "802.11 for Wi-Fi".

3. **Network Layer:** It is the 3rd layer determining the path of the data packets. This layer is responsible for routing of data packets using **IP Addressing**.
4. **Transport Layer:** It is the 4th layer that guarantees the transportation/sending of data successfully. It includes the error checking operation.
5. **Session Layer:** It is the 5th layer, identifies the established system session between different network entities. It controls dialogues between computers .For instance, while accessing a system remotely, session is created between your computer and the remote system.
6. **Presentation Layer:** It is the 6th layer that does the translation of data to the next layer (Prepare the data to the Application Layer). Encryption and decryption protocols occur in this layer such as, Secure Socket Layer (SSL).
7. **Application Layer:** It is the 7th layer, which acts as the user interface platform comprising of software within the system.

11.2.2. TCP/IP

Transmission Control Protocol/Internet Protocol, **TCP/IP** is a set of protocols which governs communications among all computers on the Internet. TCP/IP protocol tells how information should be packaged, sent, and received, as well as how to get to its destination. See Figure 11.10

TCP WORKING: TCP/IP is a combination of two protocols: Transmission Control Protocol (TCP) and Internet Protocol (IP). The Internet Protocol typically specifies the logistics of the packets that are sent out over networks; it specifies the packets which have to go, where to go and how to get there. The Transmission Control Protocol is accountable for guaranteeing the trustworthy transmission of data. It checks if any packet is not transmitted and submits it again.

Frequent TCP/IP Protocols

- **HTTP** – It is used between a web client and a web server and it guarantees non-*secure* data transmissions.

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